

ePBL Internship Report

Intelligent Machine Fault Diagnosis Using Vibration Signal Processing and Machine Learning

**Domain: Artificial Intelligence and Machine
Learning**

Submitted By

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1 Abstract

Machine fault diagnosis is an essential aspect of predictive maintenance, enabling the early detection of bearing defects and reducing equipment downtime. This project presents an intelligent machine fault diagnosis system that combines vibration signal processing with machine learning for accurate bearing fault classification. The publicly available Case Western Reserve University (CWRU) bearing dataset is used to analyze vibration signals under healthy and faulty conditions. Initially, the vibration signals are processed in the time and frequency domains using Fast Fourier Transform (FFT), followed by segmentation into fixed-size windows for statistical feature extraction. Features such as Mean, Standard Deviation, Root Mean Square (RMS), Variance, Peak, Peak-to-Peak Value, Skewness, and Kurtosis are extracted and used to train a Random Forest classifier to classify Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault conditions. Experimental results demonstrate that the proposed approach provides highly accurate fault classification with low computational complexity, making it suitable for intelligent condition monitoring and predictive maintenance applications. The proposed system offers a reliable, efficient, and scalable solution that can be extended to real-time industrial monitoring using advanced machine learning and IoT technologies.

Keywords: Predictive Maintenance, Bearing Fault Diagnosis, Vibration Signal Processing, Machine Learning, Random Forest, Fast Fourier Transform (FFT), Statistical Feature Extraction, CWRU Bearing Dataset, Industry 4.0.

2 Introduction and Objectives

The rapid advancement of Industry 4.0 has transformed traditional manufacturing into intelligent and automated production systems, where predictive maintenance plays a crucial role in ensuring equipment reliability and operational efficiency. Rotating machinery such as electric motors, pumps, compressors, turbines, and gearboxes are fundamental components in modern industries, with rolling element bearings being one of the most failure-prone mechanical elements. Bearing failures can lead to unexpected machine breakdowns, increased maintenance costs, reduced productivity, and potential safety hazards. Consequently, the development of intelligent fault diagnosis systems has become essential for early fault detection and effective condition monitoring.

Vibration signal analysis has emerged as one of the most reliable techniques for monitoring the health of rotating machinery because vibration signals contain valuable information about the operating condition of bearings. Conventional fault diagnosis methods often depend on manual inspection and expert interpretation, which are time-consuming and less efficient for large-scale industrial applications. Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the automatic analysis of vibration signals, allowing faults to be detected with higher accuracy and minimal human intervention. By extracting meaningful statistical features from vibration signals, machine learning algorithms can effectively distinguish between healthy and faulty bearing conditions, thereby improving the reliability of predictive maintenance systems.

This project presents an intelligent machine fault diagnosis system that integrates vibration signal processing with machine learning for automated bearing fault classification. The publicly available Case Western Reserve University (CWRU) Bearing Dataset is utilized to analyze vibration signals under healthy and faulty operating conditions.

Time-domain analysis, Fast Fourier Transform (FFT), statistical feature extraction, and a Random Forest classifier are employed to identify Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault conditions. The proposed approach provides an efficient, accurate, and computationally simple solution for bearing fault diagnosis, making it suitable for predictive maintenance and future real-time industrial monitoring applications.

The objectives of the project are:

- To develop an intelligent bearing fault diagnosis system.
- To analyze vibration signals using time and frequency domain techniques.
- To extract statistical features for accurate fault identification.
- To classify bearing faults using a Random Forest machine learning model.

3 Methodology

3.1 Dataset & Signal Preprocessing

The proposed fault diagnosis system utilizes the Case Western Reserve University (CWRU) Bearing Dataset, one of the most widely used benchmark datasets for bearing fault diagnosis research. The dataset contains vibration signals collected from a motor-driven bearing test rig operating under different fault conditions and load variations. In this project, four bearing conditions were considered: Healthy Bearing, Ball Fault, Inner Race Fault, and Outer Race Fault. The vibration signals were acquired from the Drive End (DE) accelerometer with a sampling frequency of 12 kHz, providing high-resolution measurements suitable for fault analysis.

The raw vibration signals are continuous time-series data and cannot be directly used for machine learning algorithms. Therefore, signal preprocessing was performed before feature extraction and classification. Initially, the vibration signals were visualized in the time domain to analyze their amplitude characteristics. Frequency-domain analysis was then carried out using the Fast Fourier Transform (FFT) to identify the dominant frequency components associated with different bearing fault conditions. To generate sufficient training samples, each vibration signal was segmented into non-overlapping windows of 1024 samples. This segmentation produced a total of 594 feature vectors, enabling efficient training and evaluation of the machine learning model while preserving the local vibration characteristics of each fault condition.

For each segmented window, statistical features including Mean, Standard Deviation, Root Mean Square (RMS), Variance, Peak Value, Peak-to-Peak Value, Skewness, and Kurtosis were extracted. These features effectively summarize the vibration characteristics while significantly reducing the dimensionality of the original signal. The resulting feature vectors were labeled according to their respective bearing conditions and used to train a Random Forest classifier for automatic fault classification. Compared with deep learning approaches, the proposed feature-based machine learning method offers lower computational complexity, improved interpretability, and excellent classification performance, making it suitable for predictive maintenance and real-time industrial condition monitoring applications.

3.2 System Architecture

An intelligent machine fault diagnosis system was developed in Python using a structured machine learning pipeline for automated bearing fault classification. The proposed architecture consists of the following stages:

1. Data Acquisition: The vibration signals were obtained from the Case Western Reserve University (CWRU) Bearing Dataset, which contains vibration data for Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault conditions. The dataset was imported into the Python environment using the SciPy and NumPy libraries for efficient signal processing.

2. Signal Processing and Feature Extraction: The raw vibration signals were initially analyzed in the time domain to examine their amplitude characteristics. Fast Fourier Transform (FFT) was then applied to obtain the frequency spectrum of the signals. Each vibration signal was segmented into 1024-sample non-overlapping windows, and statistical features including Mean, Standard Deviation, Root Mean Square (RMS), Variance, Peak Value, Peak-to-Peak Value, Skewness, and Kurtosis were extracted.

3. Machine Learning Classification: A Random Forest Classifier with 100 decision trees, implemented using the Scikit-learn library, was selected as the classification model. The extracted feature vectors were divided into training and testing datasets using an 80:20 train-test split. The Random Forest algorithm was chosen for its high accuracy, robustness to overfitting, and ability to model complex relationships within vibration data.

4. Performance Evaluation: The performance of the proposed model was evaluated using Accuracy, Precision, Recall, F1-Score, Classification Report, Confusion Matrix, and Feature Importance Analysis. These evaluation metrics provide a comprehensive assessment of the effectiveness and reliability of the intelligent bearing fault diagnosis system.

4 Implementation Code

4.1 Importing Required Libraries: The required Python libraries were imported to perform vibration signal processing, feature extraction, visualization, and machine learning classification. NumPy and Pandas were used for numerical computation and data manipulation, SciPy was used to load MATLAB (.mat) files and perform statistical analysis, Matplotlib was used for visualization, and Scikit-learn was used for implementing the Random Forest classifier.

```
1 import os
2 import glob
3 import numpy as np
4 import pandas as pd
5 import scipy.io
6 from scipy import stats
7 import matplotlib.pyplot as plt
```

```

8
9 from sklearn.model_selection import train_test_split
10 from sklearn.ensemble import RandomForestClassifier
11 from sklearn.metrics import classification_report
12 from sklearn.metrics import confusion_matrix
13 from sklearn.metrics import ConfusionMatrixDisplay

```

4.2 Loading the CWRU Bearing Dataset: The vibration signals were loaded from the Case Western Reserve University (CWRU) Bearing Dataset. MATLAB (.mat) files corresponding to Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault conditions were imported using the SciPy library.

```

1 healthy = scipy.io.loadmat("97_Normal_0.mat")
2 ball = scipy.io.loadmat("118_0.mat")
3 inner = scipy.io.loadmat("105_0.mat")
4 outer = scipy.io.loadmat("130@6_0.mat")
5
6 healthy_signal = healthy["X097_DE_time"].flatten()
7 ball_signal = ball["X118_DE_time"].flatten()
8 inner_signal = inner["X105_DE_time"].flatten()
9 outer_signal = outer["X130_DE_time"].flatten()

```

4.3 Time-Domain Signal Visualization: The vibration signals were visualized in the time domain to observe the amplitude variations and compare the characteristics of healthy and faulty bearing conditions.

```

1 plt.figure(figsize=(10,4))
2 plt.plot(healthy_signal[:4096])
3 plt.xlabel("Samples")
4 plt.ylabel("Amplitude")
5 plt.title("Healthy Bearing Vibration Signal")
6 plt.grid(True)
7 plt.show()

```

4.4 Fast Fourier Transform (FFT): Fast Fourier Transform (FFT) was applied to convert the vibration signal from the time domain into the frequency domain for identifying dominant frequency components associated with bearing faults.

```

1 fs = 12000
2
3 fft = np.fft.fft(healthy_signal[:4096])
4 freq = np.fft.fftfreq(len(healthy_signal[:4096]), d=1/fs)
5
6 positive = freq >= 0
7
8 plt.plot(freq[positive], np.abs(fft)[positive])
9 plt.xlabel("Frequency (Hz)")
10 plt.ylabel("Magnitude")
11 plt.title("FFT Spectrum")
12 plt.grid(True)

```

```
13 plt.show()
```

4.5 Statistical Feature Extraction: Each vibration signal was divided into non-overlapping windows of 1024 samples, and statistical features were extracted from every segment. These features represent the bearing condition while reducing the dimensionality of the original vibration signal.

```
1 def extract_features(signal):
2
3     mean = np.mean(signal)
4     std = np.std(signal)
5     rms = np.sqrt(np.mean(signal**2))
6     variance = np.var(signal)
7     peak = np.max(np.abs(signal))
8     p2p = np.ptp(signal)
9     skewness = stats.skew(signal)
10    kurtosis = stats.kurtosis(signal)
11
12    return [mean, std, rms, variance,
13            peak, p2p, skewness, kurtosis]
```

4.6 Dataset Preparation: The extracted statistical features from all vibration signal segments were combined to create the final dataset. Each feature vector was assigned a corresponding class label representing the bearing condition.

```
1 df = pd.DataFrame(features)
2
3 df.columns = [
4     "Mean",
5     "Std",
6     "RMS",
7     "Variance",
8     "Peak",
9     "Peak_to_Peak",
10    "Skewness",
11    "Kurtosis"
12 ]
13
14 df["Class"] = labels
```

4.7 Random Forest Classification: A Random Forest classifier consisting of 100 decision trees was used to classify the extracted statistical features into four bearing conditions.

```
1 X = df.drop("Class", axis=1)
2
3 y = df["Class"]
4
5 X_train, X_test, y_train, y_test = train_test_split(
6     X,
```

```

7     y,
8     test_size=0.2,
9     random_state=42,
10    stratify=y
11 )
12
13 rf = RandomForestClassifier(
14     n_estimators=100,
15     random_state=42
16 )
17
18 rf.fit(X_train,y_train)
19
20 y_pred = rf.predict(X_test)

```

4.8 Performance Evaluation: The trained model was evaluated using standard machine learning metrics including Precision, Recall, F1-Score, Accuracy, and Confusion Matrix.

```

1 print(classification_report(
2     y_test,
3     y_pred
4 ))
5
6 cm = confusion_matrix(
7     y_test,
8     y_pred
9 )
10
11 ConfusionMatrixDisplay(
12     confusion_matrix=cm
13 ).plot()

```

4.9 Feature Importance Analysis: Feature importance analysis was performed to identify the statistical features that contributed most significantly to bearing fault classification.

```

1 importance = pd.Series(
2     rf.feature_importances_,
3     index=X.columns
4 )
5
6 importance.sort_values().plot(
7     kind="barh"
8 )
9
10 plt.xlabel("Importance")
11 plt.ylabel("Features")
12 plt.show()

```

5 Results

This chapter presents the experimental results obtained from the proposed intelligent machine fault diagnosis system. The vibration signals were processed using time-domain and frequency-domain analysis, followed by statistical feature extraction and Random Forest classification. The effectiveness of the proposed model was evaluated using standard machine learning performance metrics.

5.1 Time-Domain Vibration Signal: The healthy bearing vibration signal was analyzed in the time domain to examine its amplitude characteristics. The waveform exhibits a stable vibration pattern, representing normal operating conditions and serving as a reference for fault analysis.

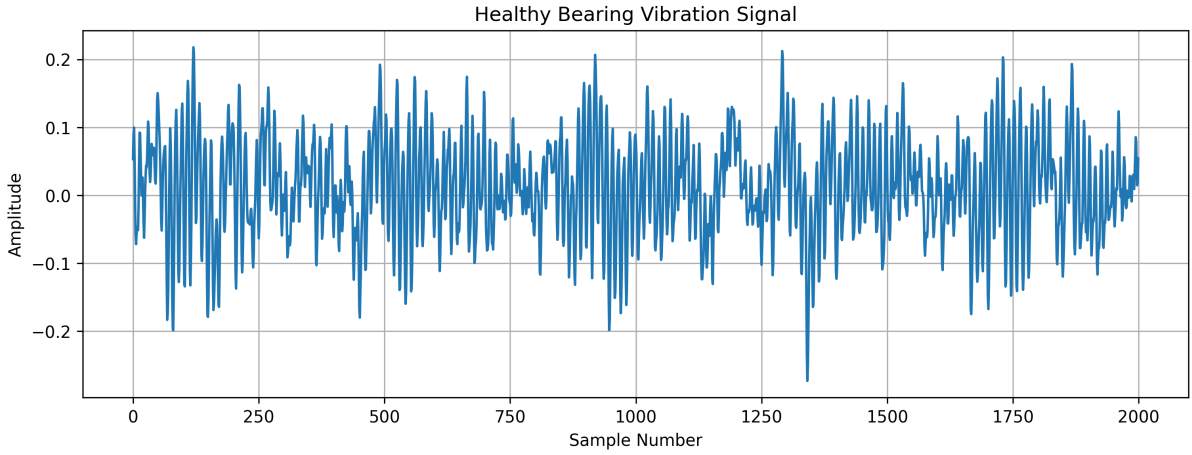


Figure 1: Healthy Bearing Vibration Signal.

5.2 Comparison of Bearing Vibration Signals: The vibration signals of Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault conditions were compared. Faulty bearings show noticeable variations in amplitude and waveform patterns compared to the healthy bearing, indicating the presence of mechanical defects.

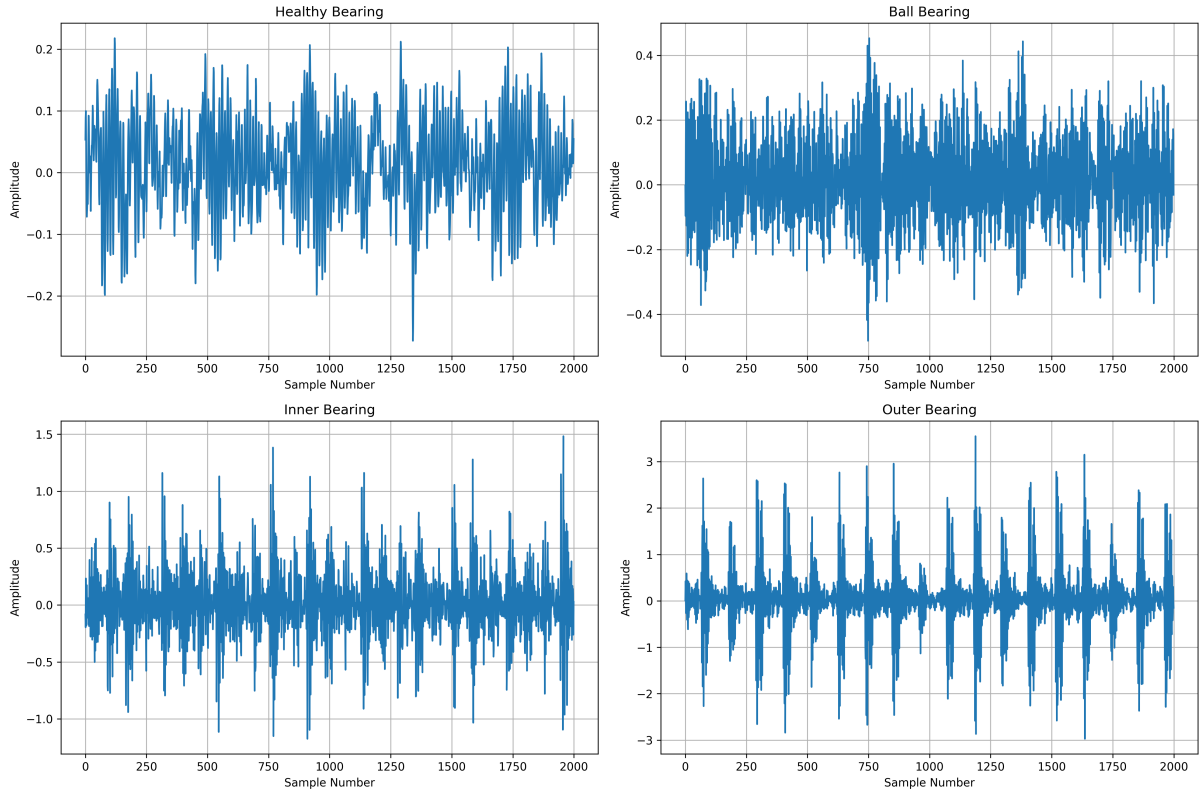


Figure 2: Comparison of Healthy and Faulty Bearing Signals.

5.3 Frequency Spectrum Analysis: Fast Fourier Transform (FFT) was applied to obtain the frequency spectrum of the vibration signal. The spectrum highlights the dominant frequency components, providing additional information for fault identification.

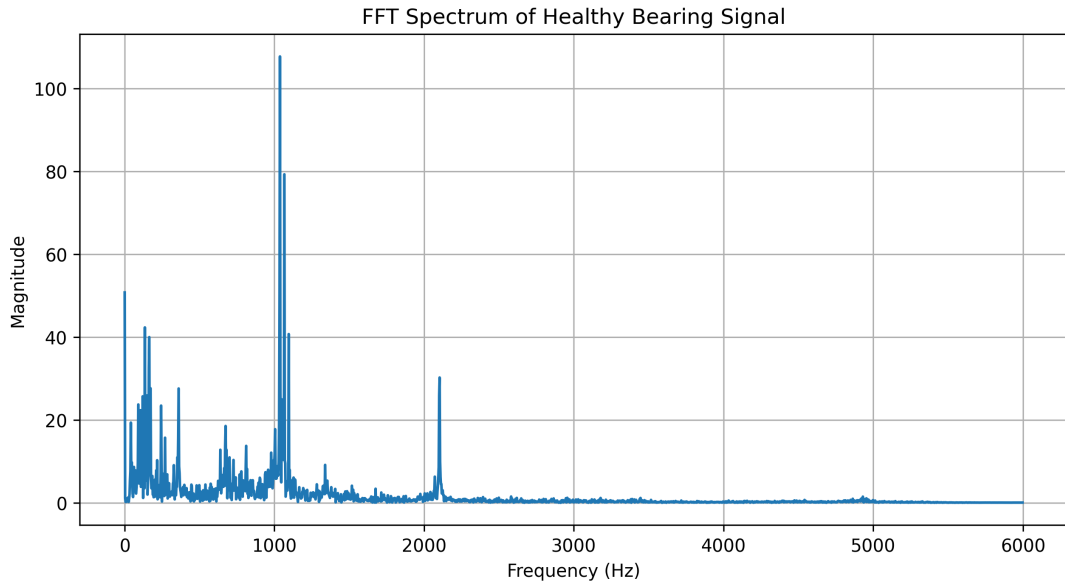


Figure 3: FFT Spectrum of Healthy Bearing Signal.

5.4 Statistical Feature Extraction: Eight statistical features—Mean, Standard Deviation, RMS, Variance, Peak, Peak-to-Peak, Skewness, and Kurtosis—were extracted

from each signal segment. These features effectively represent the vibration characteristics while reducing the dimensionality of the original signal.

	Mean	Std	RMS	Variance	Peak	Peak_to_Peak	Skewness	Kurtosis	Class
0	0.012421	0.074960	0.075982	0.005619	0.217794	0.416396	-0.162976	-0.243664	Healthy
1	0.011589	0.069465	0.070425	0.004825	0.272869	0.485031	-0.107386	0.184817	Healthy
2	0.014643	0.076996	0.078377	0.005928	0.212370	0.420569	0.050329	-0.376182	Healthy
3	0.010937	0.078934	0.079688	0.006231	0.226765	0.449775	-0.163588	-0.034564	Healthy
4	0.013359	0.074922	0.076104	0.005613	0.188588	0.375508	-0.028448	-0.321609	Healthy

Figure 4: Statistical Features Extracted from Vibration Signals.

5.5 Classification Performance: The extracted features were used to train the Random Forest classifier. The classification report demonstrates excellent performance, with high Precision, Recall, and F1-Score for all bearing conditions.

	precision	recall	f1-score	support
Ball	1.00	1.00	1.00	24
Healthy	1.00	1.00	1.00	48
Inner	1.00	1.00	1.00	23
Outer	1.00	1.00	1.00	24
accuracy			1.00	119
macro avg	1.00	1.00	1.00	119
weighted avg	1.00	1.00	1.00	119

Figure 5: Classification Report of the Random Forest Classifier.

5.6 Confusion Matrix: The confusion matrix illustrates the classification performance of the proposed model. Most test samples were correctly classified, indicating the reliability and effectiveness of the developed fault diagnosis system.

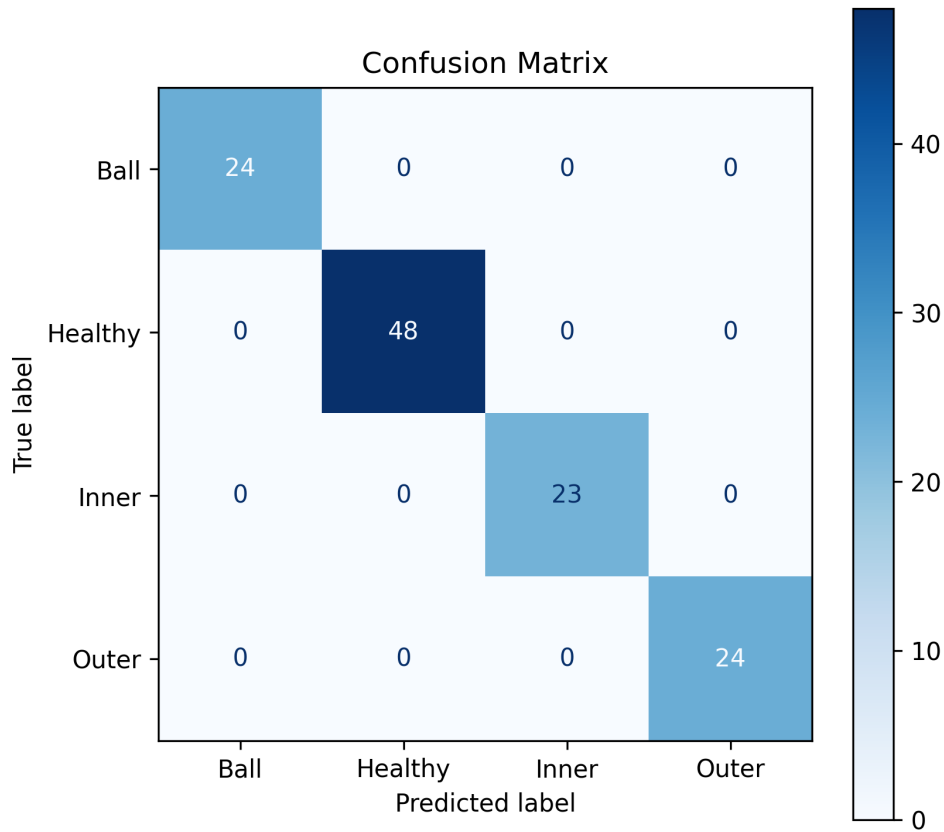


Figure 6: Confusion Matrix of the Random Forest Classifier.

5.7 Feature Importance Analysis: Feature importance analysis was performed to determine the contribution of each statistical feature to bearing fault classification. RMS, Standard Deviation, and Peak-to-Peak Value were identified as the most significant features, while Skewness showed the lowest contribution.

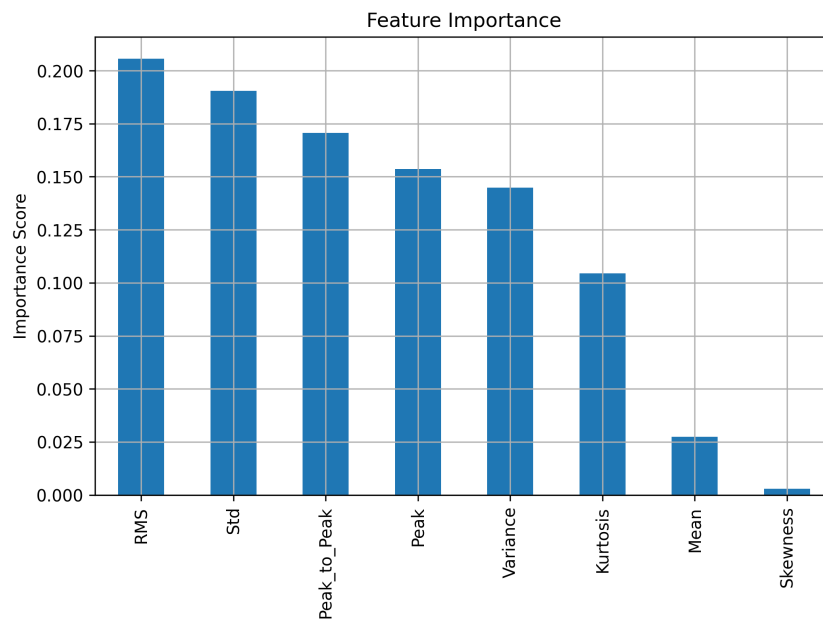
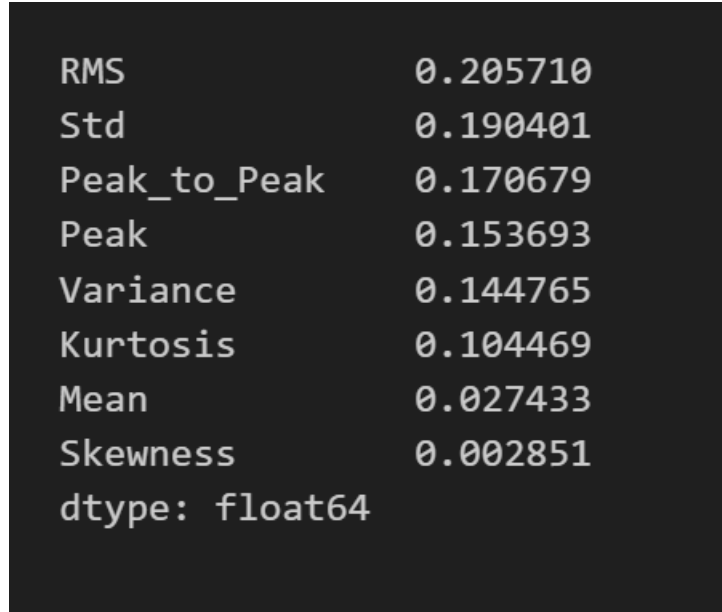


Figure 7: Feature Importance Analysis.



RMS	0.205710
Std	0.190401
Peak_to_Peak	0.170679
Peak	0.153693
Variance	0.144765
Kurtosis	0.104469
Mean	0.027433
Skewness	0.002851
dtype: float64	

Figure 8: Ranking of Statistical Features Based on Importance Scores.

Discussion: The experimental results demonstrate that the proposed vibration signal processing and machine learning approach effectively classifies Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault conditions. Statistical feature extraction combined with the Random Forest classifier provides accurate fault identification with low computational complexity, making the proposed system suitable for predictive maintenance and industrial condition monitoring applications.

6 Conclusion and Future Enhancements

This project successfully developed an intelligent machine fault diagnosis system using vibration signal processing and machine learning techniques for accurate bearing fault classification. The Case Western Reserve University (CWRU) Bearing Dataset was used to analyze vibration signals under four operating conditions: Healthy, Ball Fault, Inner Race Fault, and Outer Race Fault. The vibration signals were processed using time-domain analysis and Fast Fourier Transform (FFT), followed by the extraction of statistical features, including Mean, Standard Deviation, RMS, Variance, Peak, Peak-to-Peak, Skewness, and Kurtosis. These features effectively captured the characteristics of different bearing conditions while reducing the complexity of the original vibration signals. A Random Forest classifier was then trained using the extracted feature set and demonstrated excellent classification performance. The experimental results confirm that the proposed methodology provides an accurate, reliable, and computationally efficient solution for intelligent bearing fault diagnosis. Overall, the developed system offers a practical approach for predictive maintenance and industrial condition monitoring, with the potential to reduce unexpected machine failures and improve equipment reliability.

Future enhancements include:

- Implement real-time fault monitoring using IoT sensors.
- Explore deep learning models such as CNN and LSTM.
- Extend the system to diagnose additional machine faults.
- Deploy the model on embedded and edge devices.
- Improve accuracy using multi-sensor data fusion.

References

- [1] K. A. Loparo, “Case Western Reserve University Bearing Data Center,” Case Western Reserve University. [Online]. Available: <https://engineering.case.edu/bearingdatacenter>
- [2] S. Lessmeier, J. K. Kimotho, D. Zimmer, and G. Sextro, “Condition Monitoring of Bearing Damage in Electromechanical Drive Systems by Using Machine Learning Methods,” *IEEE International Conference on Prognostics and Health Management (ICPHM)*, 2016.
- [3] S. J. Pan and Q. Yang, “A Survey on Transfer Learning,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [4] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] R. B. Randall, *Vibration-Based Condition Monitoring: Industrial, Aerospace and Automotive Applications*. John Wiley & Sons, 2011.